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Bose–Einstein Condensation in the Large Deviations Regime with Applications to Information System Models

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Abstract. We study the large deviations behavior of systems that admit a certain form of a product distribution, which is frequently encountered both in Physics and in various information system models. First, to fix ideas, we demonstrate a simple calculation of the large deviations rate function for a single constraint (event). Under certain conditions, the behavior of this function is shown to exhibit an analogue of Bose–Einstein condensation (BEC). More interestingly, we also study the large deviations rate function associated with two constraints (and the extension to any number of constraints is conceptually straightforward). The phase diagram of this rate function is shown to exhibit as many as seven phases, and it suggests a two–dimensional generalization of the notion of BEC (or more generally, a multi–dimensional BEC). While the results are illustrated for a simple model, the underlying principles are actually rather general. We also discuss several applications and implications pertaining to information system models.

1. Introduction

While the theory of statistical physics is traditionally concerned with typical or almost typical events, the closely related theory of large deviations deals with rare events whose probabilities are exponentially small in the size of the system. More precisely, large deviations theory is concerned with the exponential decay rate of probabilities of certain rare events, as the number of observations grows without bound. In statistical mechanics, there has always been some interest in the statistics of rare events (the Kramers escape problem being one example [1]). More recently, the interest in rare events has grown due to several applications. For example, in many cases it is important to know the probability of an extinction event in non–equilibrium models of epidemics [2, 3]. Another example is the measurement of fluctuation theorems [4, 5], such as the Jarzynski equality, which rely on probing rare events. Interest in rare events has also emerged recently in the statistics of records and other stochastic processes [6, 7, 8]. Finally, the calculation of large deviations is a natural framework within which one can define non–equilibrium free energy analogues [9, 10, 11, 12].

In this paper, we consider large deviations pertaining to product measures. In particular, we focus on the probability that some quantity of relevance would exceed a certain threshold. The paper contains two parts. In the first, we give a discussion with a tutorial flavor, which focuses on the calculation of the probability of a simple single large deviations events. Our aim, in this part, is to point out, for a non–expert reader, two important aspects: The first is the relation between large deviations theory and conventional statistical physics, and the second is the fact that phase transitions can be observed in the large deviations regime even in simple systems with no interactions, where phase transitions are not expected in the usual regime, of analyzing the typical behavior of the system. In particular, for product distributions of a certain form, a direct analogue of Bose–Einstein condensation (BEC) can be observed. Following this tutorial part, we turn to the second part of the paper, where we present results that extend the calculations of a single event to accommodate two simultaneous events, and the further extension to any finite and fixed number of events is then conceptually obvious. We show that even when the two events are physically closely related, the phase diagram can exhibit as many as seven different phases. This means that the large deviations point–of–view actually suggests a multi–dimensional extension of the notion of BEC. Furthermore, we compare phase diagrams of large deviations rate functions pertaining to inequality events to those of equality events and it turns out that these two phase diagrams are very different.

To fix ideas, we first illustrate the results for a simple hopping model (closely related to the models studied in [13, 14, 15]), but they remain valid for fairly general forms of product distributions, and as such, apply to many physical systems and information system models. Examples of these range from black–body radiation (for a related calculation of large deviations in ideal quantum gases, see [16]), zero-range processes (in and out of equilibrium) [17], Jackson networks, which emerge in queuing theory

(and which are essentially analogous to zero range processes, but with no conservation of the particle number) [18, 19], driven–diffusive systems [20], and many others. These product distributions also arise in additional engineering applications. For example, this is the natural distribution for a one–way Markov chain, which is defined by an ordered set of states, where the only allowed transitions from each state are the self–transition and a transition to the next state. One–way Markov processes are commonly used in statistical modeling for a wide spectrum of application areas, including information theory, communications and signal processing (see Section 5 for details).

The outline of the remaining part of this paper is as follows: In Section 2, we illustrate our results, without the detailed derivation, using a simple one–dimensional hopping model, which may describe transport in a disordered medium. In Section 3, we derive general results for the large deviations rate function of a single constraint. In Section 4, we extend the derivation to incorporate two constraints, and then display the corresponding phase diagram. finally, in Section 5, we discuss several applications to information system models.

2. Informal Illustration of the Results

Throughout this paper, we consider systems whose steady-state behavior admits a probability distribution of the product form

$$\mathrm{P}(\{n_i\}) = \frac{1}{Z}\prod_i p_i^{n_i}, \tag{1}$$

where n_i is the number of "particles" in "lattice site" i of the system and Z is a normalization constant. This means that $\{n_i\}$ are independent geometric random variables with parameters $\{p_i\}$. The immediate relevance of this model is the distribution of the occupation numbers $\{n_i\}$ of the various energy levels in the grand canonical ensemble of an ideal boson gas, where $p_i = ze^{-\beta\epsilon_i}$, z being the fugacity, β – the inverse temperature, and $\{\epsilon_i\}$ are the corresponding energy levels. Other natural applications of this model, which were mentioned briefly in the Introduction, will be reviewed in detail in Section 5. One can easily generalize our results to the case where each factor in the product of (1) is $p_i^{n_i}/n_i^b$. For the sake of simplicity, however, we confine ourselves throughout to the form (1), for which $b = 0$. For concreteness and intuition, we focus on a particularly simple dynamical model with such a steady–state distribution (for related models, see [13, 14, 15]). The model is defined on a one–dimensional lattice, with M sites, labeled by $i = 0, 1, \ldots, M-1$. A configuration of the system is defined by the number of particles $n_i = 0, 1, \ldots, M-1$ at each site. The evolution is governed by random sequential dynamics defined by the following rules: Particles enter into the system via site $i = 0$ at rate α (there is no exclusion between the particles). If at site i $n_i > 0$, a particle is transferred from site i to site $i+1$ at rate μ_i. At site $i = M-1$, particles leave the lattice at rate μ_{M-1}. The model, as illustrated in Fig. 1, is therefore non–conserving only at the edges of the system. It can be considered as a simple model for transport in a disordered medium or, more pictorially, as a model of customers being

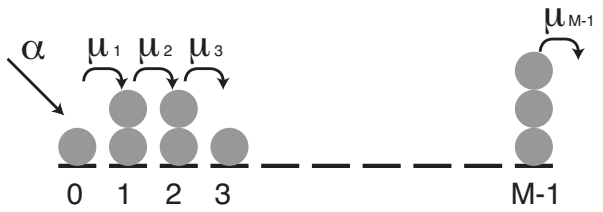


Figure 1. An illustration of the hopping model. Particles enter into the system from the left at rate α. If site i is occupied, a particle is transferred to site $i+1$ at rate μ_i. Particles leave the system at rate μ_{M-1} on the right–hand side. For a closely related model see, for example, [13].

served along a sequence of M consecutive queues, from left to right. In the latter case, each site represents a server. In the realm of queuing network theory, this model is a specific example of a Jackson network [18, 19], and in steady–state, it admits a product of distributions of geometric random variables, provided that the rate at which particles flow into the system is small enough, so that the system does not overflow, namely, in this case, $\alpha < \min_i\{\mu_i\}$. Specifically, the steady–state probability of a configuration $(n_0, \ldots, n_{M-1})$, in this example, is given by

$$\mathrm{P}(n_0, n_1, \ldots, n_{M-1}) = \frac{1}{Z} \prod_{i=0}^{M-1} \left(\frac{\alpha}{\mu_i}\right)^{n_i}, \tag{2}$$

which falls in the framework of (1) with $p_i = \alpha/\mu_i$.

Our interest is in calculating the probability of a certain large–deviations (rare) event $\mathbf{X}$. A simple example of such an event, customarily considered in large deviations theory, is that the total number of particles in the lattice exceeds some threshold, that is, $\mathbf{X} = \{(n_0, n_1, \ldots, n_{M-1}) : \quad \sum_{i=0}^{M-1} n_i \geq N\}$. Consider the thermodynamic limit where both N and M grow without bound, such that their ratio $N/M \equiv U$ is kept fixed. If U exceeds a minimum value, given by its average value, and which we shall denote $U_{\min}$, this event becomes asymptotically rare, and its probability, $\Pr\{\mathbf{X}\}$ decays with M asymptotically exponentially at the same as $\exp[-M \cdot J(U)]$, where $J(U)$ is referred to as the *large deviations rate function* in large deviations theory. Our interest will therefore be primarily in the evaluation of $J(U)$. Another relevant question would be about characterizing those configurations of the system that dominate $J(U)$. In other words, given that the event $\mathbf{X}$ has occurred, what are the system configurations that one is likely to observe?

In general, $J(U)$ may not be a smooth function. It may exhibit singularities (e.g., discontinuities in the derivatives of $J(\cdot)$) at some value $U = U_c$ (perhaps even at more than one such value). In the sequel, these will be referred to as *phase transitions* of the large deviations rate function. These phase transitions may be manifested not merely in possible singularities of the function J, but more interestingly, in condensation phenomena pertaining to the dominant configurations of the large deviations event in

question. For example, under certain conditions on the asymptotic behavior of rates $\{\mu_i\}$ (analogous to conditions on the density of states in BEC), for $U > U_c$, the dominant configurations become condensed: Although the total number of sites M grows without bound, a macroscopic fraction of the particles reside only in one of them. Loosely speaking, the particles are essentially jammed at the site (or server) with the slowest exit rate [20]. The value U_c is analogous to the critical density in the ordinary BEC transition. For large deviations events of the type $\sum_i n_i \geq N$, the rate function exhibits an additional phase transition: When $U < U_{\min}$, the event in question is no longer rare and so $J(U) = 0$. This is a direct result of looking at an event defined by an inequality constraint rather than an equality constraint. If one considers instead a constraint of the form $\sum_i n_i = N$, one would find two phases only, a condensed phase ($U > U_c$) and a non–condensed phase ($U < U_c$), and not three. In the sequel, we will elaborate more on this difference between equality constraints and inequality constraints.

Interestingly, condensation phenomena occur also for other constraints defined in terms of various linear combinations of $\{n_i\}$. For example, $\hat{T} \equiv \sum_i n_i/\mu_i$ is a plausible estimate for the total time that a particle would spend in the system, because n_i/μ_i is the expected time that each particle spends at site i before being moved, in its turn, to the right. Consider now the event $\hat{T} \geq M \cdot V$. The large deviations behavior of this event also exhibits two phase transitions, one at $V = V_{\min}$, where $J(V)$ ceases to be identically zero and becomes strictly positive, and the other at $V = V_c$, from the non–condensed to the condensed phase, with V_c depending on the rates in the system. Once again, in the condensed phase, particles essentially jam at the site with the smallest exit rate.

More surprising and interesting is the phase diagram obtained for the joint probability of two rare events pertaining to two different linear combinations of $\{n_i\}$, say, $\Pr\{\sum_i n_i \geq M \cdot U,\ \sum_i n_i/\mu_i \geq M \cdot V\}$, which decays exponentially according to $\exp\{-M \cdot J(U,V)\}$ for some rate function $J(U,V)$. As we show in the sequel, even if the two constraints are physically closely related, the phase diagram of the large deviations rate function $J(U,V)$ has a very rich phase diagram with as many as seven different phases. We find three distinct types condensed phases: one for each one of the individual events and a third one for their combination, which gives rise to the notion of a *two–dimensional condensation*. Furthermore, the phase diagram associated with the corresponding equality constraints, $\sum_i n_i = MU$ and $\sum_i n_i/\mu_i = MV$, is dramatically different from that of the inequality constraints, with two phases only rather than seven. Note, that this two dimensional condensation is very different in nature from that considered in the context of two, distinct, conserved quantities [21, 22, 23]. The two–constraint problem, in its general form, is the focus of the main part of the paper. In the next section, we present a detailed derivation of the results for the single constraint problem.

3. A Single Constraint

As mentioned in the Introduction, we begin with a simple single constraint, assuming that one has a product distribution of the form of eq. (1). Referring to the terminology of particles and sites, from the example of Section 2, consider first the probability of the event that the number of particles in the system is larger than some threshold, $\sum_i n_i \geq M \cdot U$. The large deviations evaluation of this probability is typically done using the Chernoff bound. Specifically, consider the following chain of inequalities:

$$\begin{aligned}
\Pr\left\{\sum_i n_i \geq MU\right\} &\leq \left\langle z^{\sum_i n_i - MU}\right\rangle \qquad z \geq 1 \\
&= z^{-MU}\prod_i \frac{1-p_i}{1-zp_i} \\
&= \exp\left\{-M\left[U\ln z - \frac{1}{M}\sum_i \ln\left(\frac{1-p_i}{1-zp_i}\right)\right]\right\}, \qquad (3)
\end{aligned}$$

where the angular brackets denote an expectation with respect to the distribution of eq. (1). The tightest bound, which gives the large-deviations rate function, is obtained by minimization of the Chernoff bound over z, or equivalently, by maximization of the bracketed expression at the exponent:

$$J(U) = \sup_{z\geq 1}\left[U\ln z - \lim_{M\to\infty}\frac{1}{M}\sum_{i=0}^{M-1}\ln\left(\frac{1-p_i}{1-zp_i}\right)\right], \qquad (4)$$

provided that the limit exists for all $z \geq 1$.

Note that the *Chernoff parameter* z, that undergoes optimization, is almost equivalent to the fugacity z of the grand–canonical ensemble, which controls the expected number of particles in the system, and the minimization of the bound is parallel to the usual saddle–point evaluation pertaining to the grand partition function. The only difference comes about since the Chernoff bound is concerned with the probability of the inequality event $\sum_i n_i \geq MU$, as opposed to the event $\sum_i n_i = MU$, which defines the canonical ensemble with $N = MU$. This implies that one is interested in $z \geq 1$ and when the number of particles is below its average value, $J = 0$. For rare events (with $J > 0$), the distinction between $\Pr\{\sum_i n_i \geq N\}$ and $\Pr\{\sum_i n_i = N\}$ becomes meaningless in the limit of large N due to the exponential decay of the probability with M. With this analogy, clearly, in the limit of large M (or equivalently N), the bound gives an asymptotically exact value of the rate function J (see, e.g., [24]). In other words, the calculation of the large deviations probability of a rare event is essentially identical to a change of ensembles in traditional statistical physics, with the rate function J playing the role of a free energy. An extra, somewhat trivial, phase occurs due to the constraint taking the form of an inequality and not an equality of the form $\sum_i n_i = N$. In the latter case, the phase with $J = 0$ would not exist. With this in mind, what follows in the next paragraph is standard.

As mentioned earlier, in order to proceed from eq. (4), we must assume that the limit in eq. (4) exists. We will assume that there exists a density function $g(t) \geq 0$,

integrating to unity, such that in the limit of $M \to \infty$, the fraction of $\{p_i\}$ that fall between t and $t + \mathrm{d}t$, tends to $g(t)\mathrm{d}t$ for all $t \in (0,1)$. Performing a saddle–point approximation on eq. (3) gives the following equation for the optimum choice of z:

$$U = \lim_{M\to\infty} \frac{1}{M} \sum_{i=0}^{M-1} \frac{zp_i}{1-zp_i} = \int_0^{p_m} \frac{ztg(t)\mathrm{d}t}{1-zt}, \tag{5}$$

where $p_m = \max_i p_i$, and where it is assumed that p_m is attained by the same i (say, $i = 0$ without loss of generality) for all M.‡ Let us denote

$$\boldsymbol{U}(z) \equiv z \int_0^{p_m} \frac{tg(t)\mathrm{d}t}{1-zt}. \tag{6}$$

We therefore have to solve the equation $U = \boldsymbol{U}(z)$, where the solution z is sought in the range $[1, 1/p_m)$. Now, in analogy with BEC, if $g(p_m) = 0$ and $\lim_{t\uparrow p_m} g(t)/(p_m - t)^\chi$ is positive and finite for some $\chi > 0$, then $\boldsymbol{U}(1/p_m) < \infty$, and so, the large deviations behavior exhibits a condensation. In other words, as long as U is below the the critical density:

$$U_c = \boldsymbol{U}(1/p_m) \equiv \int_0^{p_m} \frac{tg(t)\mathrm{d}t}{p_m - t}, \tag{7}$$

there is no condensation, while for $U > U_c$, condensation takes place. This means that the large deviations event in question is dominated by realizations for which n_0/N is about $U - U_c > 0$, while all other states have negligible relative contributions. Here n_0 is the occupation at the site $i = 0$, corresponding to p_m. Denoting by $U_{\min} = \boldsymbol{U}(1)$, the average particle density in the system, the corresponding large deviations rate function is given by:

$$J(U) = \begin{cases} 0 & U < U_{\min} \\ U\ln z - \int_0^{p_m} \mathrm{d}t g(t) \ln\left(\frac{1-t}{1-zt}\right) & U_{\min} \le U < U_c \\ U\ln\left(\frac{1}{p_m}\right) - \int_0^{p_m} \mathrm{d}t g(t) \ln\left(\frac{1-t}{1-t/p_m}\right) & U \ge U_c \end{cases}$$

so that $\Pr\{\sum_i n_i \ge N\}$ is of the exponential order of $\exp\{-MJ(U)\}$ and the large deviations rate function exhibits three phases: the first is where U is below the average value, the second is the non-condensed phase, and the third is the condensed phase.

The above derivation can be extended quite straightforwardly to deal with more general large deviations events, defined in terms of arbitrary linear combinations of $\{n_i\}$, that is, events of the form $\{(n_0, n_1, \ldots, n_{M-1}) : \ \sum_{i=0}^{M-1} u_i n_i \ge M \cdot U\}$, where $\{u_i\}_{i=0}^{M-1}$ are arbitrary deterministic constants. For a meaningful definition of the asymptotic regime, one has to define the behavior of the infinite sequence $u_0, u_1, u_2, \ldots$, as was done concerning the infinite sequence of parameters $p_0, p_1, p_2, \ldots$. For the sake of simplicity, we will assume u_i to be a function of p_i, i.e., $u_i = u(p_i)$, for a certain given function $u : [0,1] \to \mathbb{R}$. We are then considering large deviations events of the form $\sum_{i=0}^{M-1} n_i u(p_i) \ge MU$. In the example discussed in Section 2, $u(p) = p/\alpha$, so that the sum becomes $\sum_{i=0}^{M-1} n_i/\mu_i \ge MU$. In the example of the ideal Bose gas, where $p_i = e^{-\beta\epsilon_i}$,

‡ Note that for $p_i = e^{-\beta\epsilon_i}$, this is exactly the classical equation that underlies the BEC.

the energy constraint $\sum_i n_i\epsilon_i > MU$ corresponds to $u(p) = -\frac{1}{\beta}\ln p$ §. Similarly as in the earlier derivation, the saddle–point equation is now given by

$$U = \int_0^{p_m} \frac{tu(t)z^{u(t)}g(t)\mathrm{d}t}{1 - tz^{u(t)}},$$

whose solution z should be sought in the interval $[1, Z)$, where $Z \equiv \inf_{p\in[0,p_m]}\{p^{-1/u(p)}\}$. Thus, for

$$U > U_c \equiv \int_0^{p_m} \frac{tu(t)Z^{u(t)}g(t)\mathrm{d}t}{1 - tZ^{u(t)}},$$

condensation takes place, provided that $U_c < \infty$.

Several comments are now in order:

(i) It is a simple exercise to show that in two dimensions and above, an ordinary black–body would undergo a condensation when a constraint on the total number of photons is considered. This is evident by identifying $g(t)$ as the density of states of the photons, U as the density of photons in the event considered and $p_i = e^{-\beta\varepsilon_i}$ with ε_i the energy of a photon in mode i.

(ii) Different constraints can lead to condensates in different places. For example, assume that the hopping rates in the model of Section 2 are ordered so that the slowest site is at site $i = 0$ and the fastest is at site $i = M - 1$. By looking at a constraint on U, one obtains a condensation at site $i = 0$. However, if one looks at a constraint on the quantity $Q = \sum_{i\le M/2}(\mu_i - \mu_{M/2})^{\psi}n_i$, one can obtain a condensation at $i = M/2$ if ψ is large enough.

(iii) In the ordinary BEC, where $u(t) \equiv 1$, the critical density could be finite only if $g(p_m) = 0$ and $\lim_{t\uparrow p_m} g(t)/(p_m - t)^{\chi}$ is positive and finite for some $\chi > 0$. In the more general case considered now, there are choices for non–negative functions $u(t)$ such that $U_c < \infty$ even if g does not vanish at p_m. What counts is the rate at which the denominator of the integrand, $1 - tz_0^{u(t)}$, tends to zero as $t \to p_m$. If $1 - tz_0^{u(t)}$ behaves like $|t - p_m|^{\chi}$ in the neighborhood of p_m, for some $0 < \chi < 1$, and $g(t)$ is continuous and finite at $t = p_m$, then $U_c < \infty$. This in turn is possible because then the corresponding $u(t)$ would behave like $\log[(1 - |t - p_m|^{\chi})/t]$, which is positive in the neighborhood of p_m.

Having covered the single constraint problem, we now turn to the more interesting case where two constraints are considered simultaneously. Note that the analogy with a change of an ensemble is much weaker here. When considering large deviations, there is a freedom to choose any combination of constraints, so that in contrast to the usual statistical physics, the phase diagrams can have arbitrary dimensions.

4. Two Constraints

Having viewed the BEC from a large deviations perspective, it is instructive to further extend the scope and consider the joint large deviations behavior of two events or more.

§ Albeit, in this case, the corresponding constraint does not give rise to condensation.

Consider the rate function of two joint events

$$\Pr\left\{\sum_{i=0}^{M-1} u_i n_i \geq MU,\ \ \sum_{i=0}^{M-1} v_i n_i \geq MV\right\},$$

where, once again, for the sake of simplicity, we assume that u_i and v_i depend on i only via p_i, i.e., $u_i = u(p_i)$ and $v_i = v(p_i)$ for certain given functions $u(\cdot)$ and $v(\cdot)$. We confine ourselves to the case where the functions $u(\cdot)$ and $v(\cdot)$ are non–negative. This accommodates the examples discussed earlier in Section 2. Denoting $\boldsymbol{X} = \{\sum_{i=0}^{M-1} u(p_i)n_i \geq MU,\ \ \sum_{i=0}^{M-1} v(p_i)n_i \geq MV\}$, and applying a two–dimensional Chernoff bound, we have:

$$\begin{aligned}
&\Pr\{\mathbf{X}\} \\
&\leq \left\langle z_1^{\sum_{i=0}^{M-1} u(p_i)n_i - MU} \cdot z_2^{\sum_{i=0}^{M-1} v(p_i)n_i - MV} \right\rangle \qquad z_1 \geq 1,\ z_2 \geq 1 \\
&= z_1^{-MU} z_2^{-MV} \prod_i \left[(1-p_i)\sum_{n_i=0}^{\infty} [p_i z_1^{u(p_i)} z_2^{v(p_i)})]^{n_i}\right] \\
&= z_1^{-MU} z_2^{-MV} \prod_i \left[\frac{1-p_i}{1-p_i z_1^{u(p_i)} z_2^{v(p_i)}}\right] \qquad \forall i\ z_1^{u(p_i)} z_2^{v(p_i)} p_i < 1 \\
&= \exp\left\{-M\left[U\ln z_1 + V \ln z_2 - \frac{1}{M}\sum_{i=0}^{M-1} \ln\left(\frac{1-p_i}{1-p_i\ z_1^{u(p_i)} z_2^{v(p_i)}}\right)\right]\right\}. \qquad (8)
\end{aligned}$$

Again, the limitation $z_1 \geq 1$ and $z_2 \geq 1$ ensures that when we look at events where U and V take on values smaller than the expectations $\lim_{M\to\infty} \frac{1}{M}\sum_i u(p_i)\langle n_i\rangle$ and $\lim_{M\to\infty} \frac{1}{M}\sum_i v(p_i)\langle n_i\rangle$, respectively, the rate function would vanish. As before, to derive the rate function, we maximize the expression in the square brackets (which is a saddle–point analysis) by equating its partial derivatives with respect to z_1 and z_2 to zero. In the thermodynamic limit, we get the following two equations with the two unknowns z_1 and z_2:

$$\begin{aligned}
U &= \boldsymbol{U}(z_1, z_2) \equiv \int_0^{p_m} \frac{t u(t) z_1^{u(t)} z_2^{v(t)} g(t)\mathrm{d}t}{1 - t z_1^{u(t)} z_2^{v(t)}} \\
V &= \boldsymbol{V}(z_1, z_2) \equiv \int_0^{p_m} \frac{t v(t) z_1^{u(t)} z_2^{v(t)} g(t)\mathrm{d}t}{1 - t z_1^{u(t)} z_2^{v(t)}} \qquad (9)
\end{aligned}$$

where as before, p_m is the maximum of $\{p_i\}$, which is again assumed to be attained at $i = 0$ for all M. In analogy to usual BEC, z_1 and z_2 are jointly limited by the inequality $\sup_t[t z_1^{u(t)} z_2^{v(t)}] < 1$, or equivalently,

$$\sup_{0\leq t\leq p_m} [u(t)\ln z_1 + v(t)\ln z_2 + \ln t] < 0. \qquad (10)$$

In the sequel, we will refer to the following notation: For a given z_1, let $\phi(z_1)$ be the supremum of the values of z_2 that do not violate eq. (10), and let $\mathcal{A} = \{(z_1, z_2) :\ z_1 \geq 1,\ z_2 \geq 1,\ z_2 < \phi(z_1)\}$. We now use the eqs. (9) and (10) to derive the phase diagram for the large deviations rate function. For convenience, the final results are summarized

towards the end of this section. The phase diagram, shown in Fig. 2, has seven different phases.

Phase 0: not a rare event. The first, trivial, phase occurs when both U and V take on values below the expectations, $\boldsymbol{U}(1,1)$ and $\boldsymbol{V}(1,1)$, respectively. This is the region where the events are not rare and so, $J(U,V)=0$.

Phase 1: no condensation. This phase is analogous to the non–condensed phase of the single event. Here, as long as the pair (U,V) falls in a region for which the equations

$$U=\boldsymbol{U}(z_1,z_2); \qquad V=\boldsymbol{V}(z_1,z_2)$$

have a solution $(z_1,z_2)\in\mathcal{A}$, then one may substitute this solution into the Chernoff bound and obtain the rate function, which in the thermodynamic limit is given by

$$J(U,V)=U\ln z_1+V\ln z_2-\int \mathrm{d}t g(t)\ln\left[\frac{1-t}{1-tz_1^{u(t)}z_2^{v(t)}}\right].$$

This phase is the image (under the transformation defined by the pair of equations $U=\boldsymbol{U}(z_1,z_2)$, $V=\boldsymbol{V}(z_1,z_2)$) of the set $\mathcal{A}$ in the z_1–z_2 plane: It is surrounded by three curves that connect the points A, B and C in Fig. 2. The curve A–B corresponds to the collection of points where $z_1=1$, while z_2 varies from 1 (point A) up to its maximum allowed value $z_2=\phi(1)\equiv Z_2$ (point B). Similarly, the curve A–C corresponds to $z_2=1$ and z_1 varying from 1 to $\phi^{-1}(1)\equiv Z_1$. Finally, the curve B–C corresponds to the curve $z_2=\phi(z_1)$, where as z_1 increases from 1 to Z_1, $\phi(z_1)$ decreases from Z_2 to 1. The image of the latter curve in the U–V plane will be denoted by $V=\Psi(U)$. Note that Fig. 2 assumes that the curve A–B is above the curve A–C, namely, that $\boldsymbol{U}(1,z_2)\geq\boldsymbol{U}(z_1,1)$ implies $\boldsymbol{V}(1,z_2)\geq\boldsymbol{V}(z_1,1)$. In the appendix, we prove that this is indeed always the case.

Phase 2: two–dimensional condensation. We now consider the regime above the curve $V=\Psi(U)$. Let us use the short–hand notation for the values that U and V take along the curve

$$\tilde{\boldsymbol{U}}(z_1)\equiv\boldsymbol{U}(z_1,\phi(z_1))$$
$$\tilde{\boldsymbol{V}}(z_1)\equiv\boldsymbol{V}(z_1,\phi(z_1)).$$

We assume, for the moment, that they are both finite for all $z_1\in[1,Z_1]$ and that p_m, the achiever of $\sup_t[tz_1^{u(t)}[\phi(z_1)]^{v(t)}]$, is independent of z_1 (see the discussion in the sequel). Both these conditions are trivially met, for example, in the model and constraints discussed in Section 2. Let (U,V) be a point above the curve $V=\Psi(U)$. The calculation of the rate function is somewhat more involved than the single constraint case. To describe it, we need to give values for both z_1 and z_2. To do this, we note that in analogy to usual BEC, we have:

$$U-\tilde{\boldsymbol{U}}(z_1)=\lim_{M\to\infty}\left[\frac{1}{M}\cdot\frac{p_m u(p_m)z_1^{u(p_m)}z_2^{v(p_m)}}{1-p_m z_1^{u(p_m)}z_2^{v(p_m)}}\right]$$

and similarly,

$$V - \tilde{\boldsymbol{V}}(z_1) = \lim_{M\to\infty}\left[\frac{1}{M}\cdot\frac{p_m v(p_m) z_1^{u(p_m)} z_2^{v(p_m)}}{1 - p_m z_1^{u(p_m)} z_2^{v(p_m)}}\right] .$$

As in the ordinary BEC, where a prescription has to be specified for how the fugacity approaches the condensation value in the condensed phase, these equations essentially give a prescription for taking the values of the fugacities to a point where $\sup_t[t z_1^{u(t)}[\phi(z_1)]^{v(t)}] = 1$, as the thermodynamics limit is taken. Using these, we see that

$$\frac{V - \tilde{\boldsymbol{V}}(z_1)}{U - \tilde{\boldsymbol{U}}(z_1)} = \frac{v(p_m)}{u(p_m)} .$$

This equation specifies, given a point (U, V) above the curve $V = \Psi(U)$, the choice of z_1, which we shall denote by z_1^*, and hence also the choice of z_2, which is $z_2^* = \phi(z_1^*)$. The large deviations event is dominated by the state corresponding to $t = p_m$. Thus, the rate function is given by

$$J(U,V) = U\ln z_1^* + V\ln z_2^* - \int g(t)\mathrm{d}t \ln\left[\frac{1-t}{1 - t(z_1^*)^{u(t)}(z_2^*)^{v(t)}}\right] .$$

It must be kept in mind, however, that this solution is not applicable to all points (U, V) above the curve $V = \Psi(U)$. To understand the limitation, it is instructive to look at the geometric interpretation of the above equation for z_1^*: The expression $[V - \tilde{\boldsymbol{V}}(z_1)]/[U - \tilde{\boldsymbol{U}}(z_1)]$ is the slope of the straight line connecting the point (U, V) to the point $(\tilde{\boldsymbol{U}}(z_1), \tilde{\boldsymbol{V}}(z_1))$ on the curve $V = \Psi(U)$, and the equation tells us that this slope must be equal to $v(p_m)/u(p_m)$, which is a given constant. Therefore, this solution is applicable only to points (U, V) above the curve $V = \Psi(U)$ which have the following property: the straight line of slope $v(p_m)/u(p_m)$ that passes through (U, V) must intersect the curve $V = \Psi(U)$ between points B and C. The set of points with this property, which corresponds to the region of two–dimensional condensation is limited by the curve $V = \Psi(U)$ (between B and C) and the two parallel straight lines of slope $v(p_m)/u(p_m)$, passing through B and C (see Fig. 2).

Phase 3: non–condensed and dominated by the U–constraint. The region below the curve A–C (see Fig. 2) is characterized by $z_2 = 1$ and $z_1 \geq 1$. The value of z_2 is fixed at unity since we are considering values of V which are below the corresponding average value conditioned on the given value of U. This means that there is a non–condensate large deviations behavior that is dominated by that of the constraint $\sum_i u(p_i)n_i \geq MU$ alone. In other words, the other event, $\sum_i v(p_i)n_i \geq MV$, has no impact. The rate function is given by minimizing the term in the square brackets in eq. (8) with $z_2 = 1$. Denoting the obtained value of z_1 by z_1^*, the rate function is given by

$$J(U,V) = U\ln z_1^* - \int_0^{p_m} \mathrm{d}t g(t) \ln\left[\frac{1-t}{1 - t(z_1^*)^{u(t)}}\right] .$$

This phase is bounded on the right by a vertical line (see Fig. 2), where the constraint $\sum_i u(p_i)n_i \geq MU$ condenses with $z_2 = 1$.

Phase 4: condensed and dominated by the U–constraint. Following the reasoning of phase 3, the region below the straight line of slope $v(p_m)/u(p_m)$, passing via C, is the corresponding condensed phase of this single event $\sum_i u(p_i)n_i \geq MU$ (one-dimensional condensation), where the constraint $\sum_i v(p_i)n_i \geq MV$ has no impact. The upper bound on the phase can be inferred by noting that on the line emerging from point C in the figure, $z_2 = 1$.

The last two phases can be inferred from a symmetry consideration, where the two constraints interchange their roles.

Phase 5: non–condensed and dominated by the V–constraint. See the discussion for phase 3.

Phase 6: condensed and dominated by the V–constraint. See the discussion for phase 4.

Let us examine now more closely the assumption that $\tilde{\boldsymbol{U}}(z_1)$ and $\tilde{\boldsymbol{V}}(z_1)$ are both finite for a continuum of values of z_1. In the two–dimensional case considered now, this issue is more involved than in the one–dimensional case: In the one dimensional case, the relevant integral, computed at the maximum allowed value of the fugacity parameter z, may be finite if the density $g(t)$ vanishes at $t = p_m$ (the achiever of $\min_p p^{-1/u(p)}$), and tends to zero sufficiently rapidly as $t \to p_m$. By contrast, in the two–dimensional case considered now, the achiever of $\sup_t\{tz_1^{u(t)}[\phi(z_1)]^{v(t)}\}$, may depend, in general, on z_1, and it is inconceivable to expect $g(t)$ to vanish at all these values of t, which may form a continuum. (In fact, if $g(t) = 0$ for an interval, then this interval has no contribution to the integrals altogether.) Nonetheless, there is a class of special cases where this situation does not arise – the cases where the maximizing value of t turns out to be independent of z_1: For example, if $u(t)$ and $v(t)$ are both monotonically non-decreasing, then $\sup_t\{tz_1^{u(t)}[\phi(z_1)]^{v(t)}\}$ is always achieved at $t = p_m$, independent of z_1, where now p_m is again the maximum value of p across the support of the density $g(t)$. In this case, as in the one–dimensional case, if $g(t) \to 0$ as $t \uparrow p_m$ sufficiently rapidly, then $\tilde{\boldsymbol{U}}(z_1)$ and $\tilde{\boldsymbol{V}}(z_1)$ are both finite, and then for large enough U and V, there is a condensation at the state corresponding to p_m, as explained above. It should be noted, however, that the non–decreasing monotonicity of $u(t)$ and $v(t)$ is only a sufficient condition for p_m to be independent of z_1, not a necessary condition. For example, ignoring our previous assumption on the positivity of $u(t)$ and $v(t)$, if $u(t) \equiv 1$ and $v(t) = -\ln t$, this is still true, although $v(t) = -\ln t$ is a decreasing function.

To summarize, we have identified seven phases in the U–V plane. Denoting

$$\mathcal{J}(z_1, z_2, U, V) = U \ln z_1 + V \ln z_2 - \int g(t)\mathrm{d}t \ln\left[\frac{1-t}{1 - tz_1^{u(t)}z_2^{v(t)}}\right],$$

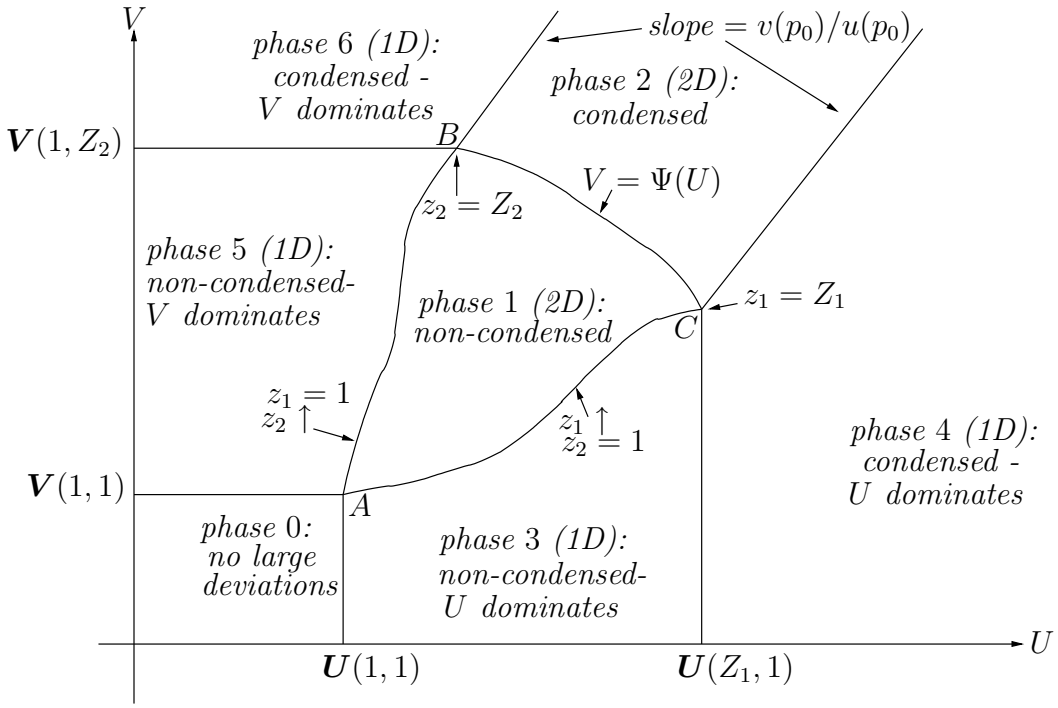


Figure 2. Phase diagram in the $U - V$ plane. Note that each one the points A, B and C is the meeting points of four different phases.

the rate function takes the following behaviors:

$$J(U,V) = \begin{cases} 0 & \text{phase 0} \\ \max_{z_1,z_2} \mathcal{J}(z_1,z_2,U,V) & \text{phase 1} \\ \mathcal{J}(z_1^*,\phi(z_1^*),U,V) & \text{phase 2} \\ \max_{z_1} \mathcal{J}(z_1,1,U,V) & \text{phase 3} \\ \mathcal{J}(Z_1,1,U,V) & \text{phase 4} \\ \max_{z_2} \mathcal{J}(1,z_2,U,V) & \text{phase 5} \\ \mathcal{J}(1,Z_2,U,V) & \text{phase 6} \end{cases}$$

It is interesting to compare this phase diagram with the one which would be obtained by considering the equality event

$$\Pr\{\mathbf{X}\} = \Pr\left\{\sum_{i=0}^{M-1} u(p_i)n_i = MU,\ \sum_{i=0}^{M-1} v(p_i)n_i = MV\right\}.$$

In this case, the values of both z_1 and z_2 would not be restricted to be larger than 1. Therefore, all phase transitions associated with either $z_1 = 1$ or $z_2 = 1$ would disappear in this case. It is straightforward to see (similarly to the derivation of phase 1), that here we have two phases only: a condensed phase and a non–condensed phase. In the z_1–z_2 plane, the set $\mathcal{A}$ is no longer limited by the inequalities $z_1 \geq 1$ and $z_2 \geq 1$, but only the curve $z_2 = \phi(z_1)$, whose image in the U–V plane is now the *entire* curve $V = \Psi(U)$, which is no longer limited by the points B and C. The region below this curve is the

non–condensed phase and the region above the curve is condensed. The condensation is always two–dimensional in character.

Finally, it would be interesting to demonstrate that in certain situations, the condensating state may jump abruptly as we move continuously in the U–V plane. In the above discussion we made specific assumptions on the functions $g(t)$, $u(t)$, and $u(t)$. In principle, it is possible to extend the calculation to cases where the achiever of $\sup_t\{tz_1^{u(t)}[\phi(z_1)]^{v(t)}\}$ takes on any finite number of values as z_1 varies between 1 and Z_1, and and that the density g vanishes (and sufficiently rapidly) at all these values of t. An interesting scenario arises, for example, in a variation of the above example, defined by the choices $u(t) \equiv 1$ and $v(t) = -\alpha - \ln t$, where $0 < \alpha < -\ln p_m$, and where as before, p_m is the maximum of t across the support of $g(t)$. In this case, it is easy to see that the achiever of $\sup_t\{tz_1^{u(t)}[\phi(z_1)]^{v(t)}\}$ is given by p_m for $z_2 < e$ ($z_1 > e^\alpha$), and by p_∞, which is the minimum of t across the support of $g(t)$, for $z_2 > e$ ($z_1 < e^\alpha$). In other words, the condensing state jumps from p_m to the other extreme, p_∞, at the point $z_1 = e^\alpha$ along the curve $V = \Psi(U)$. In this case, the two–dimensional condensed phase splits into three sub–phases. If we denote by D the point corresponding to $z_1 = e^\alpha$ along the curve $V = \Psi(U)$, then above this curve, we see three different types of two–dimensional condensation (see Fig. 3):

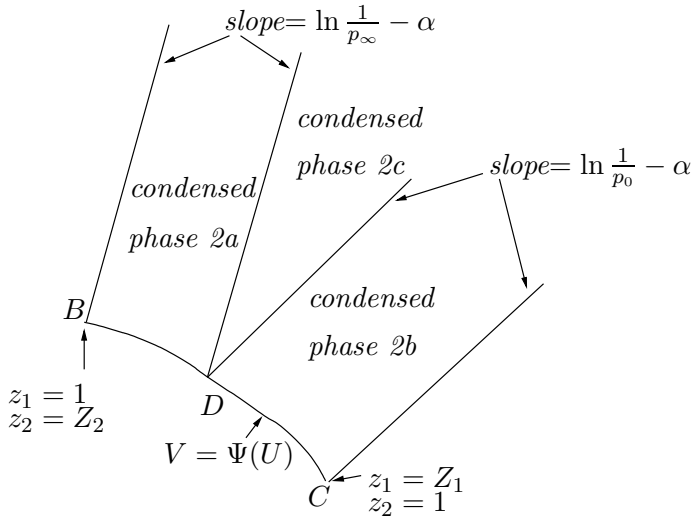


Figure 3. Zoom–in on the two-dimensional condensed phase in the example of $u(t) \equiv 1$ and $v(t) = -\alpha - \ln t$.

(i) The region limited by the curve B–D and two parallel straight lines with slope $\ln(1/p_\infty) - \alpha$, passing through points B and D (phase 2a).

(ii) The region limited by the curve D–C and two parallel straight lines with slope $\ln(1/p_0) - \alpha$, passing through points D and C (phase 2b).

(iii) The region in between 1 and 2 (phase 2c). The rate function for all points in phase 2c is the same as in the point D.

5. Applications

Our large deviations analysis focuses on events associated with linear combinations pertaining to sequences of independent (but not necessarily identically distributed) geometric random variables. Beyond the obvious relevance of this model to the grand–canonical ensemble of the ideal boson gas, as was mentioned earlier, there are quite a few additional applications, which cover, not only the realm of statistical physics, but also that of information engineering models. We mentioned briefly some of these applications in the Introduction. In this section, we discuss them in somewhat more detail.

The first application example is that of a *one–way Markov chain* (a.k.a. *left–to–right* Markov chain, in the literature of speech signal processing). A one–way Markov chain is defined by an ordered set of states $(0, 1, 2, \ldots)$, where the only allowed transitions from each state i are the self–transition $(i \to i)$ – with probability p_i, and a transition to the next state $(i \to i+1)$ – with probability $1 - p_i$, $i = 0, 1, 2, \ldots$ (see Fig. 4). Clearly, every sequence generated by a one–way Markov chain, as defined, is composed of n_0 self–transitions of state 0, followed by n_1 self–transitions of state 1, followed in turn by n_2 self–transitions of state 2, and so on, where $n_0, n_1, n_2, \ldots$ are independent, geometric random variables with parameters $\{p_i\}$.

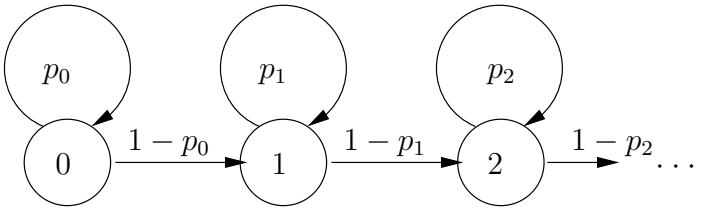


Figure 4. State transition diagram for a one–way Markov chain.

Therefore, it is clear that this model falls within our framework. The one–way Markov chain is a very useful model in a variety of application areas of information system models. A few examples are hidden Markov modeling of speech signals (see, e.g., [25] and references therein), the segmentation of signals, such as those that govern the evolution of the fading process of a communication channel (or channels that "heat up" [26]), the segmentation of electrocardiographic signals (see, e.g., [27]), beat tracking in audio signals (see, e.g., [29]), and even handwritten text recognition [27].

The interest in the large deviations behavior of linear combinations of $\{n_i\}$ is not difficult to justify, in the context of one–way Markov chains. Consider the problem of lossless data compression of the sequence of random variables $n_0, n_1, \ldots$. An elementary result in Information Theory (see, e.g., [28]) tells that the optimum code length (in bits) of the compressed version of each n_i is given by

$$\ell_i(n_i) = -\log P(n_i) = -\log[(1-p_i)p_i^{n_i}] = n_i \log(1/p_i) + \log[1/(1-p_i)],$$

which is an affine function of n_i. The large deviations event $\sum_i \ell_i(n_i) \geq N = MU$ is the

event that the total code length would exceed the limit of N. If N designates the size of a buffer in which the compressed data is stored (in order to monitor the bit rate), then this event has the meaning of a buffer overflow, whose consequence is that information is lost. We would like, of course, to keep the probability of such an event as small as possible.

Another application where independent geometric random variables naturally arise, is in queuing theory. An M/M/1 queue (see, e.g., [38]) is a common model of a queue according to which the arrivals of customers is a Poisson process of rate λ, the service is based on the principle of first come – first served (FCFS), and the service time for each customer is distributed exponentially with rate μ. As long as $\lambda < \mu$, the queue is stable (does not diverge) and the steady–state distribution of the number of customers in the queue is geometric with parameter $p = \lambda/\mu$, which is called the *utilization* of the queue. Jackson's theorem [18] extends this to an open network (a.k.a. a Jackson network) of M queues, which means that: (i) any external arrival to any given node is a Poisson process, (ii) a customer completing service at queue i either joins another queue j with probability p_{ij} or leaves the system with probability $1 - \sum_j p_{ij}$, which is non–zero for at least one queue, and (iii) all utilization parameters p_i are less than 1. Jackson's theorem tells that the steady–state joint probability distribution of the queue lengths is given by a product of individual geometric distributions with parameters $\{p_i\}$. A special case of a queuing network was considered in Section 2.

In the context of queuing networks, BEC means that one of the queues, the one with the highest utilization, becomes responsible for a bottleneck (or a traffic jam) – a linear fraction of the total number of customers spend their time in that queue due to the inefficient performance of the server of this queue relative to the arrival rate. When applied to queuing networks, our large deviations results mean that we identified BEC in an open (Jackson) network and in addition, we have characterized the rate function, as well as the phase transitions associated with it. Moreover, since we are allowing large deviations events pertaining to arbitrary linear combinations of $\{n_i\}$, one natural application example, as already discussed, is the large deviations behavior of $\sum_i n_i/\mu_i$ (with μ_i being the rate through queue no. i), which is a reasonable estimate of the total waiting time for a customer who visits all queues.

There are, of course, other network models that are known to admit a product–form steady–state distribution. One of them is the closed–network version of the Jackson network, called the *Gordon–Newell network* [39],[40]. The only difference between the Gordon–Newell network and the Jackson network is that the former is a closed network (unlike a Jackson network which is open), i.e., there is no external supply of customers and no departures from the system, and so, the total number of customers is fixed. The steady–state distribution for the Gordon–Newell network is exactly analogous to the canonical Bose–Einstein distribution, and hence it exhibits BEC under certain conditions, as was observed already in earlier work, cf. e.g., [36] and [31].

The Gordon–Newell theorem appears to be a special case of results concerning product forms of steady–state distributions in classes of models, such as the zero–

range process (ZRP) (see, e.g., [20], [17] and references therein), that are studied in the statistical physics literature. According to the ZRP model, particles (customers) that lie in an array of sites (a lattice, or more generally, the nodes of a certain graph), may hop from one site (queue) to another, and may pile up, according to certain rules (see, e.g., the example discussed in Section 2). Jackson's theorem, however, does not seem to be directly derivable as a special case since it pertains to an open network. A subsequent paper by Jackson [19] allows state–dependent service times and it seems to include the ZRP model as a special case.

Acknowledgment

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Appendix

In this appendix, we prove that $\boldsymbol{U}(1,z_2) \geq \boldsymbol{U}(z_1,1)$ implies $\boldsymbol{V}(1,z_2) \geq \boldsymbol{V}(z_1,1)$, which means that the A–B curve in Fig. 2 lies above the A–C curve.

Consider the function

$$f(x) = \frac{tg(t)e^x}{1-te^x},$$

where t is a parameter taking values in the range where the denominator is strictly positive. For a given $t \geq 0$, this function is clearly monotonically non–decreasing in x. Therefore, for all t:

$$[u(t)\ln z_1 - v(t)\ln z_2]\cdot[f(u(t)\ln z_1) - f(v(t)\ln z_2)] \geq 0.$$

Integrating over t, we get:

$$\begin{aligned}
0 \leq & \int \mathrm{d}t[u(t)\ln z_1 - v(t)\ln z_2]\cdot[f(u(t)\ln z_1) - f(v(t)\ln z_2)] \\
= & \left[\int \frac{tu(t)z_1^{u(t)}g(t)\mathrm{d}t}{1-tz_1^{u(t)}} - \int \frac{tu(t)z_2^{v(t)}g(t)\mathrm{d}t}{1-tz_2^{v(t)}}\right]\cdot\ln z_1 + \\
& \left[\int \frac{tv(t)z_2^{v(t)}g(t)\mathrm{d}t}{1-tz_2^{v(t)}} - \int \frac{tv(t)z_1^{u(t)}g(t)\mathrm{d}t}{1-tz_1^{u(t)}}\right]\cdot\ln z_2 \\
= & [\boldsymbol{U}(z_1,1) - \boldsymbol{U}(1,z_2)]\cdot\ln z_1 + [\boldsymbol{V}(1,z_2) - \boldsymbol{V}(z_1,1)]\cdot\ln z_2. \qquad (11)
\end{aligned}$$

Since the first bracketed term of the last expression is non–positive (by hypothesis) and since $\ln z_1 \geq 0$ and $\ln z_2 \geq 0$, the second bracketed term must be non–negative, which proves the argument.

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